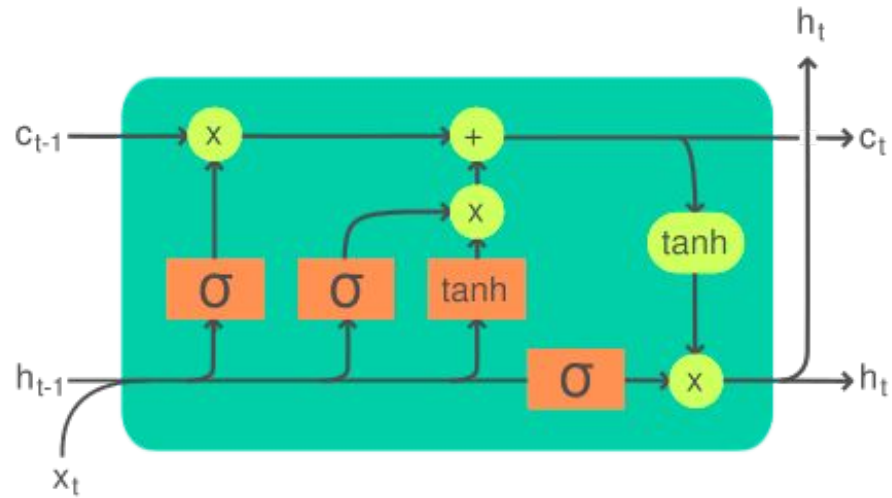


Computation Graph of Vanilla LSTM

In-Class Task: Try to draw the complete graph



Legend:

Layer



Componentwise



Copy



Concatenate



Some Python pseudocode

```
2
3 def lstm_step(x_t, h_prev, c_prev, params):
4     """Forward pass for a single timestep of an LSTM."""
5     # Params includes all weights and biases
6     W_f, U_f, b_f = params['W_f'], params['U_f'], params['b_f'] # Forget gate
7     W_i, U_i, b_i = params['W_i'], params['U_i'], params['b_i'] # Input gate
8     W_c, U_c, b_c = params['W_c'], params['U_c'], params['b_c'] # Candidate gate
9     W_o, U_o, b_o = params['W_o'], params['U_o'], params['b_o'] # Output gate
10
11     # Forget gate
12     f_t = sigmoid(W_f @ x_t + U_f @ h_prev + b_f) # First "+" is vector concatenation, second one is element-wise addition
13
14     # Input gate
15     i_t = sigmoid(W_i @ x_t + U_i @ h_prev + b_i) # First "+" is vector concatenation, second one is element-wise addition
16
17     # Candidate cell state
18     c_tilde = tanh(W_c @ x_t + U_c @ h_prev + b_c) # First "+" is vector concatenation, second one is element-wise addition
19
20     # New cell state
21     c_t = f_t * c_prev + i_t * c_tilde # "+" is element-wise addition
22
23     # Output gate
24     o_t = sigmoid(W_o @ x_t + U_o @ h_prev + b_o) # First "+" is vector concatenation, second one is element-wise addition
25
26     # New hidden state
27     h_t = o_t * tanh(c_t)
28
29     return h_t, c_t
30
31
32 cell_state = [0, 0, 0]
33 hidden_state = [0, 0, 0]
34 for x in inputs:
35     hidden_state, cell_state = lstm_step(x, hidden_state, cell_state, lstm_params)
36
```

Goldberg's Vanilla LSTM Formulation

$$s_j = R_{\text{LSTM}}(s_{j-1}, x_j) = [c_j; h_j]$$

$$c_j = f \odot c_{j-1} + i \odot z$$

$$h_j = o \odot \tanh(c_j)$$

$$i = \sigma(x_j W^{xi} + h_{j-1} W^{hi})$$

$$f = \sigma(x_j W^{xf} + h_{j-1} W^{hf})$$

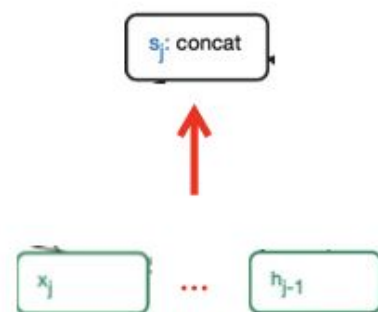
$$o = \sigma(x_j W^{xo} + h_{j-1} W^{ho})$$

$$z = \tanh(x_j W^{xz} + h_{j-1} W^{hz})$$

$$y_j = O_{\text{LSTM}}(s_j) = h_j$$

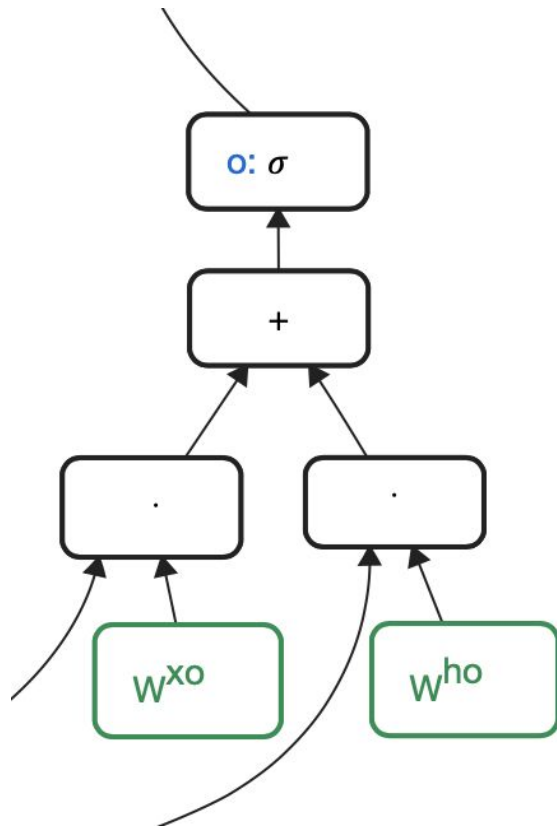
In-Class-Task

Draw a computation graph (and label the nodes as named in the formula accordingly) for producing the result of step s_j



Help

- Parameters and inputs are leaf nodes
- Some operator nodes will have labels from the formulas (named expressions)
- E.g. part of the computation graph for the output gate o
- [Chris Olah's Blog Post](#)



Some more help!

